

Background and Motivation

I am a tritium scientist at Type One Energy, working within the Infinity II fusion power plant program. My primary focus is on developing the tritium extraction system for our solid breeder blanket and supporting development of the coolant purification system.

I have used FESTIM to build pebble-scale tritium release models, varying key parameters to understand release behavior. My Python background is strong, developing physics-based simulation tools using GEKKO, NumPy, and SciPy, including both residence time models and equation orientated models. Whilst I do not have a wealth of experience with FENiCs, as part of my preparation for this program I am planning further engagement. I am applying to the Summer 2026 cohort to develop deeper FESTIM capability in direct support of Infinity II, our Fusion Power Plant, and with outcomes that will feed into both a hydrogen laboratory program currently under development and any future tritium systems built as part of the TES development.

Proposed Fellowship Project

The project addresses a central challenge in the design of the Tritium Extraction System for a fusion power plant: connecting pebble-scale tritium release physics to purge-loop system behavior, and ultimately to plant-level fuel cycle models. The work is structured in three linked stages, each of which generates direct value for our fusion power plant, Infinity II, design decisions and for establishing the simulation and experimental infrastructure to support them. The first stage refines an existing FESTIM model of tritium release from lithium ceramics. The goal is a robust, reproducible simulation setup that accurately captures transient tritium inventory and release rate as a function of temperature, sweep gas conditions, and irradiation history. Where parameter gaps exist, I want to work with the development team to identify best-practice approaches to uncertainty handling and to understand how simulation outputs can be meaningfully constrained by and compared against experimental measurements. This connects directly to experimental capability being developed through our hydrogen laboratory program, where characterization rigs are being designed specifically to measure tritium-relevant transport properties in solid breeder and structural materials. The pebble model will be benchmarked against experimental release data being developed.

The second stage integrates pebble model outputs into a system-level model of the helium purge loop TES. This model will be a conventional sweep gas circuit with getter bed extraction as starting basis, with this developing the skills to model the more commercially sensitive systems being developed. The model will capture tritium partial pressure evolution along the purge loop, extraction efficiency as a function of operating conditions, and the effect of pebble

bed packing and network geometry on bulk release behavior. Cycling simulations will assess transient performance of the pebble. Results will feed directly into Infinity II TES design trade studies.

The third stage is a stretch goal, using the pebble and system models to develop a surrogate representation of TES behavior suitable for integration into plant-scale fuel cycle frameworks, specifically a FactorySim-style dynamic model of the tritium plant such as one in PathSim as per Dark *et al.* (In Press, 2026). A central learning objective for this stage is understanding best practice for constructing FESTIM-derived surrogates that preserve the relevant physics while remaining computationally tractable at plant timescales. This capability will become increasingly important as Type One Energy moves toward a tritium facility, where validated simulation tools linking material-scale measurements to system behavior will underpin both facility design and operational safety cases.

What I Hope to Gain

The technical gaps I most want to close span three areas. The first is best-practice use of FESTIM for system-scale simulations spanning multiple components and geometries. The second is approaches to coupling FESTIM models to real experimental data. How to structure simulations so that measured transport properties from laboratory rigs can be used to constrain and validate model parameters in a rigorous and traceable way, further learning of the FESTIM teams experience with their own experimental set-ups such as LIBRA. The third is established patterns for constructing reduced-order surrogates from FESTIM for integration into wider modelling frameworks. Taken together, these capabilities will allow me and the Type One Energy team to build a coherent pipeline from laboratory measurement through component simulation to plant-level analysis.

Alignment with Fellowship Outcomes

This project delivers against two of the four listed expected outcomes: significant progress on a complex application case, in the form of a validated multi-scale simulation hierarchy from pebble to purge loop and integration of FESTIM into a broader modelling framework, via the surrogate interface to the Infinity II plant model.

Duration and Availability

I am proposing 50 % engagement over approximately 10 weeks beginning mid-June 2026, with funding through Type One Energy as my home institution. I am open to either remote or hybrid participation, with in-person time at PSFC if the program supports it. We have a

Woburn, MA office that would allow me to participate in work outside of the Fellowship while still in the Boston area.